Lab 7 – Configuring Recovery

Lab submission

Your lab submission should contain the following two files:

File 1:

1. Create a document called **Lab7_*firstname*_*lastname***, where *firstname* is your first name and *lastname* is your last name.
2. Put your full name at the top of the document.
3. Take screen captures (Use the Windows Snipping Tool) as requested in red text with the tag **<<SS##>>**, label each screen capture with the entire tag and description in red.
4. If any questions are asked, they will be labelled as **<<Q##>>** in red text as well, insert the question tag and answer into your document in order.
5. When finished submit the document, which only contains the labelled screenshots, as a .**pdf** to **Assessments>Dropbox>Lab 7 prior to the posted due date.**

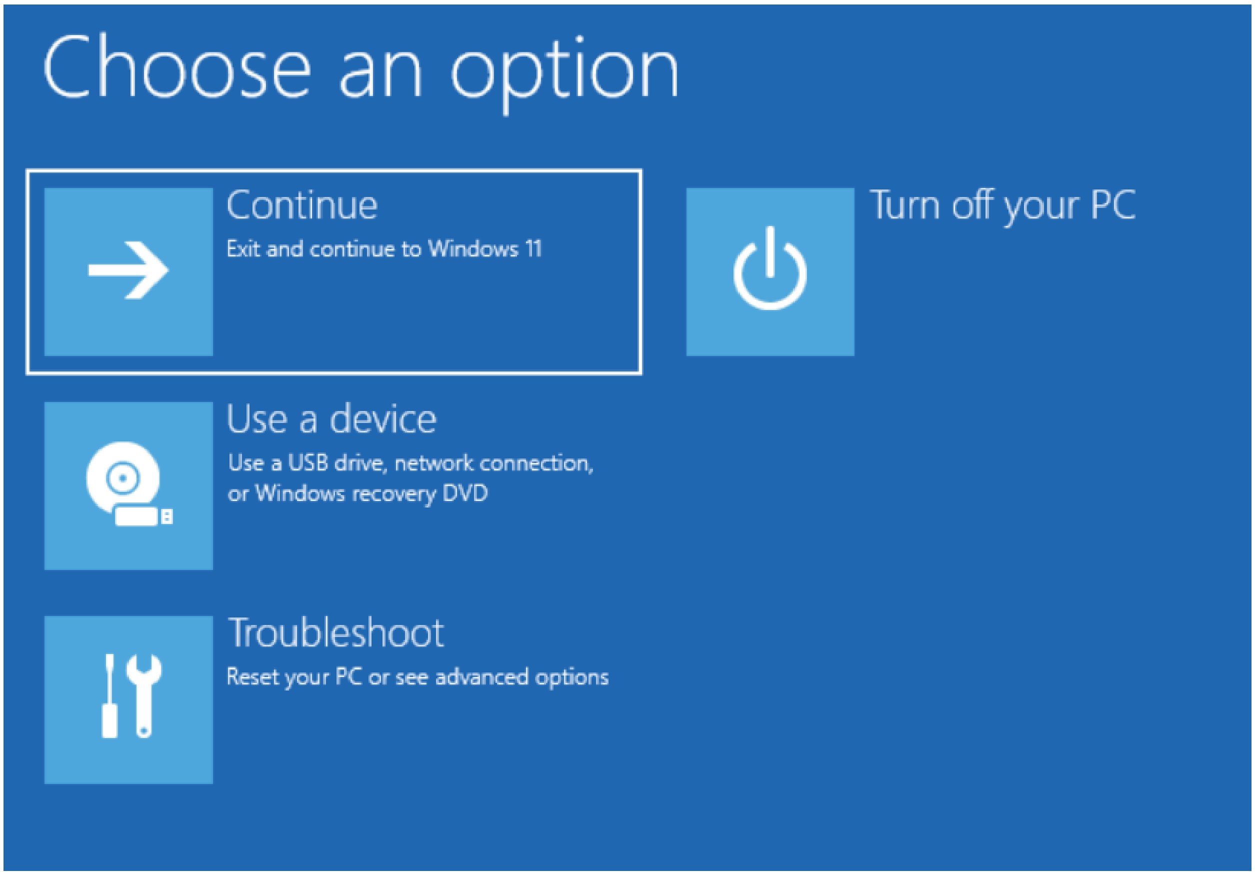
File 2:

1. Make sure each screen shot is saved as SS##, where the octothorps are the appropriate number.
2. Zip all the pictures into one file and submit to the Brightspace Dropbox.
3. Note: These pictures contain metadata which can be used for verification if any questions arise regarding academic misconduct.

A. Windows recovery and diagnostics

Boot into Advanced options

1. Power on the Windows 11 PC but do not log in.
2. Hold the Shift key while at the initial log on screen and restart the PC.
3. Eventually it will boot into a recovery screen.



This menu allows you to access important features outside of a working windows environment. This screen can also be accessed with recovery and installation media for Windows.

Continue will boot to the regular windows installation, a good option if the device had a temporary crash.

Use a Device will boot to the installation portions of recovery media.

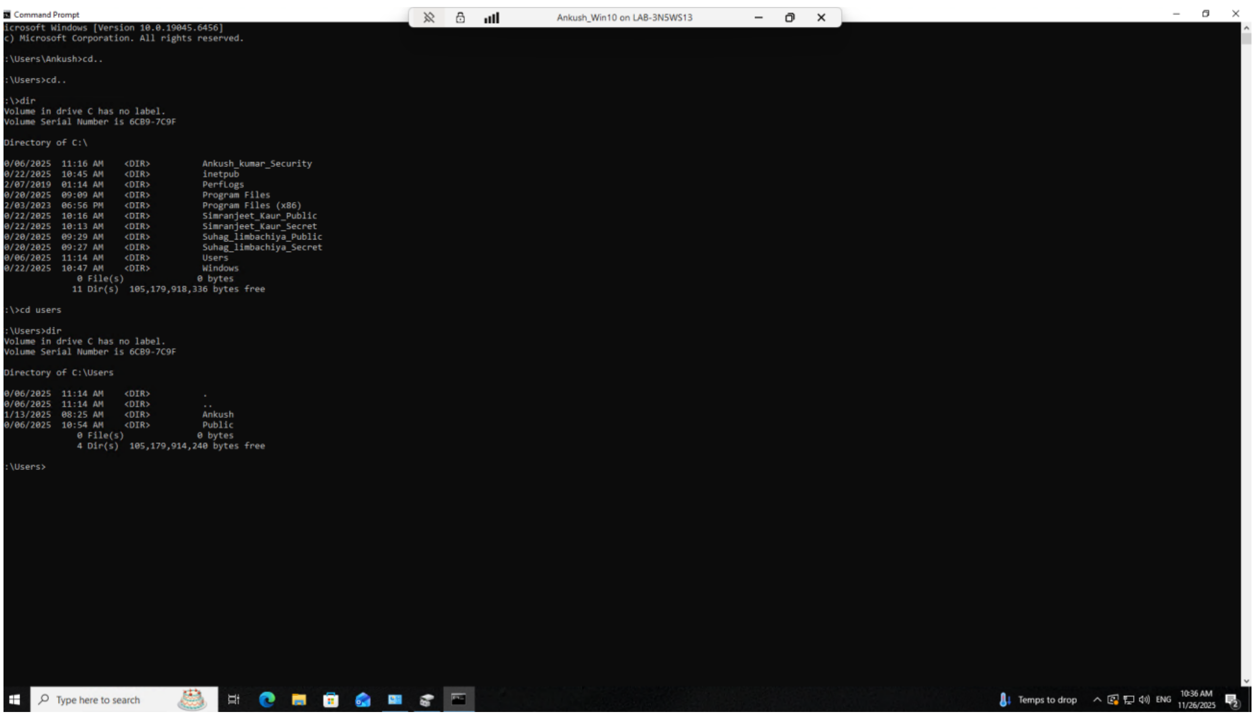
Troubleshoot will open additional recovery tools.

Command prompt in Recovery options

NOTE* Delete all volumes that were created in the previous labs (Simple, Extended, Stripped and Spanned Volumes) before continuing. You can do this from Disk Manager**

1. Select *Troubleshoot->Advanced Options*.
2. Open Command Prompt.
3. Change to the C drive by Entering the command ***C:***.
4. Enter the command ***dir***. This should list the main directory of the C: drive.
5. Enter the command ***cd users***
6. Enter the command ***dir***

<<SS01>> take a screenshot of the command prompt and its returns.



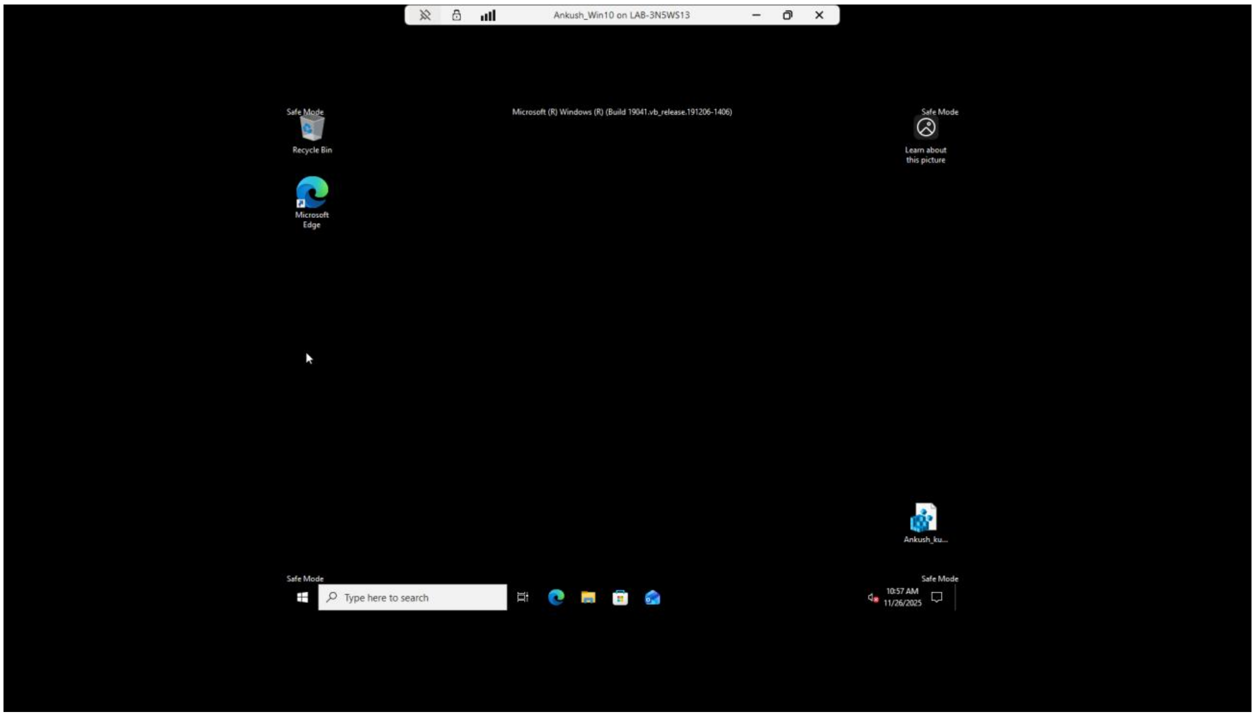
Type in exit to leave command prompt.

Boot into safe mode

1. From the main recovery window select *Troubleshoot->Advanced Options->Startup Options*.
2. Select *Restart*.
3. Select option 4 with *F4*.

The PC will reboot into Safe Mode. Safe Mode has limited functionality, by limiting services and network connectivity this mode can be used to isolate software issues.

<<SS02>> Take a screenshot of the Safe mode desktop.

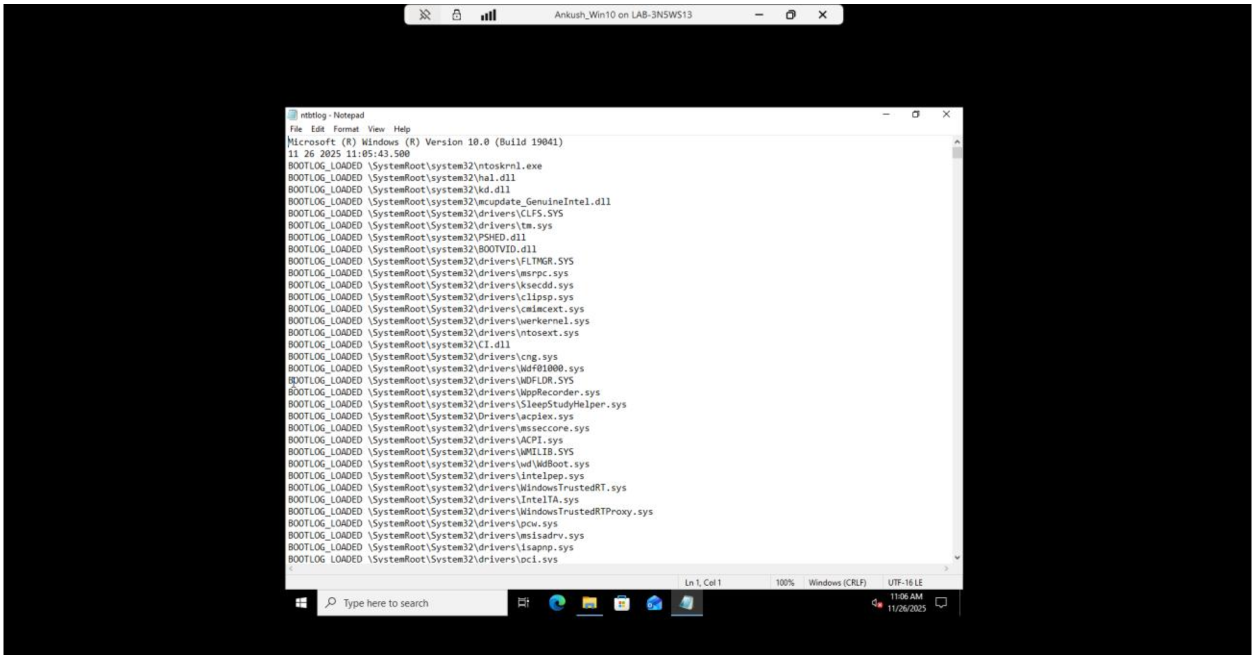


View Boot logs

1. Open File Explorer.
2. Navigate to the Windows folder.
3. Open the ntblog.

This log shows the devices, drivers, and services that were booted with the machine.

<<SS03>> Take a screenshot of the ntblog.txt document.



Power down the PC.

Create a backup

To store our backup we need more space.

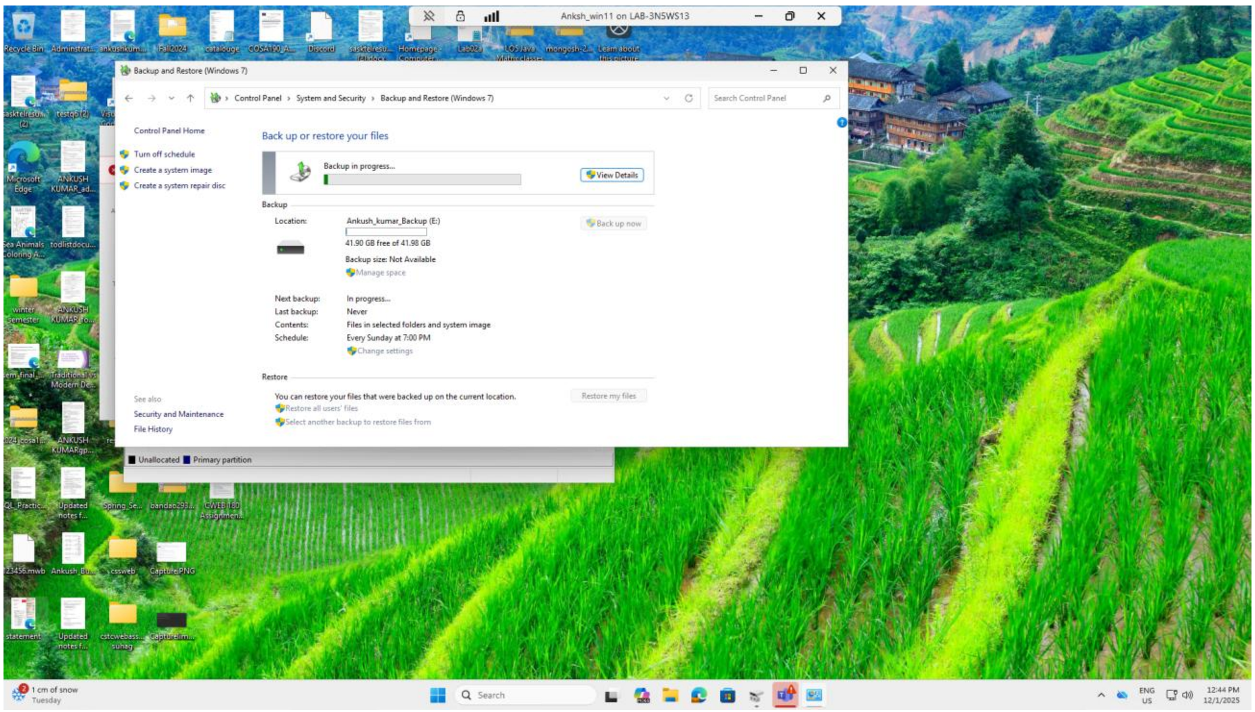
1. Create a new VHD with the following settings:
    a. **Fixed size**
    b. **42GB**
    c. Name = **Win11Backup**
2. Power on the PC.
3. Open disk manager and initialize the drive with the following settings:
    a. **GPT**
    b. **Maximum disk space**
    c. Attach to **any** letter
    d. **NTFS** File system
    e. Volume label = ***FirstName_LastName_Backup***

Once the drive is ready, we can use it for the backup.

1. Open Control Panel and select Backup and Restore (Windows 7).
2. Select Set up *Backup*.

3. Use the newly initialized drive to store the backup.
4. Manually choose to backup only the data files for the users.
5. Click *Save settings* and run backup.

<<SS04>> Take a screenshot of the back up screen, the back doesn't need to finish prior to taking the screen shot.



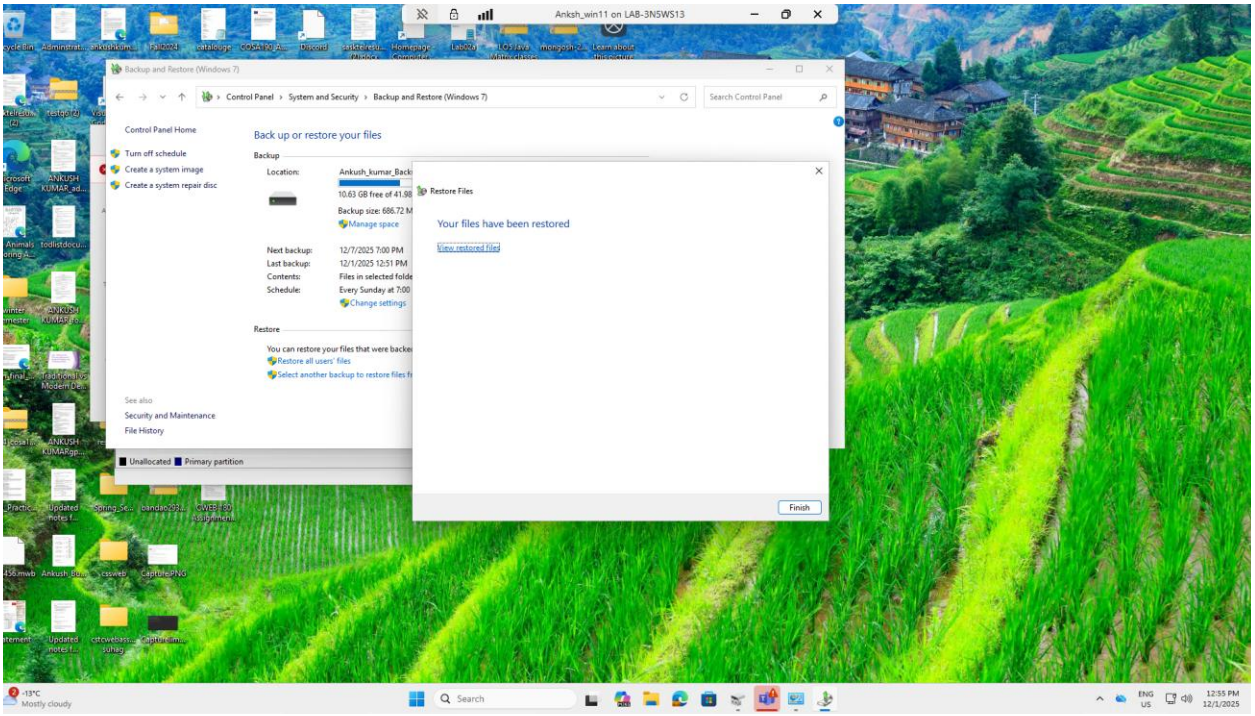
A backup is now created. It contains a disk image and would allow the user to restore a PC in the event of total failure. Caution: If the devices BIOS or hardware changes between the creating of the backup and its use, the image may not work.

Restoring files

1. Click restore my files in the bottom right corner.
2. Click Browse for files.
3. Navigate to Backup of *C:->Users->FirstInitial_LastInitial_StandardAccess->Desktop*.
4. Select the Desktop.ini file.
5. Click *Next*.
6. Restore to the original location.
7. Click *Restore*.

8. Replace the file at the destination.

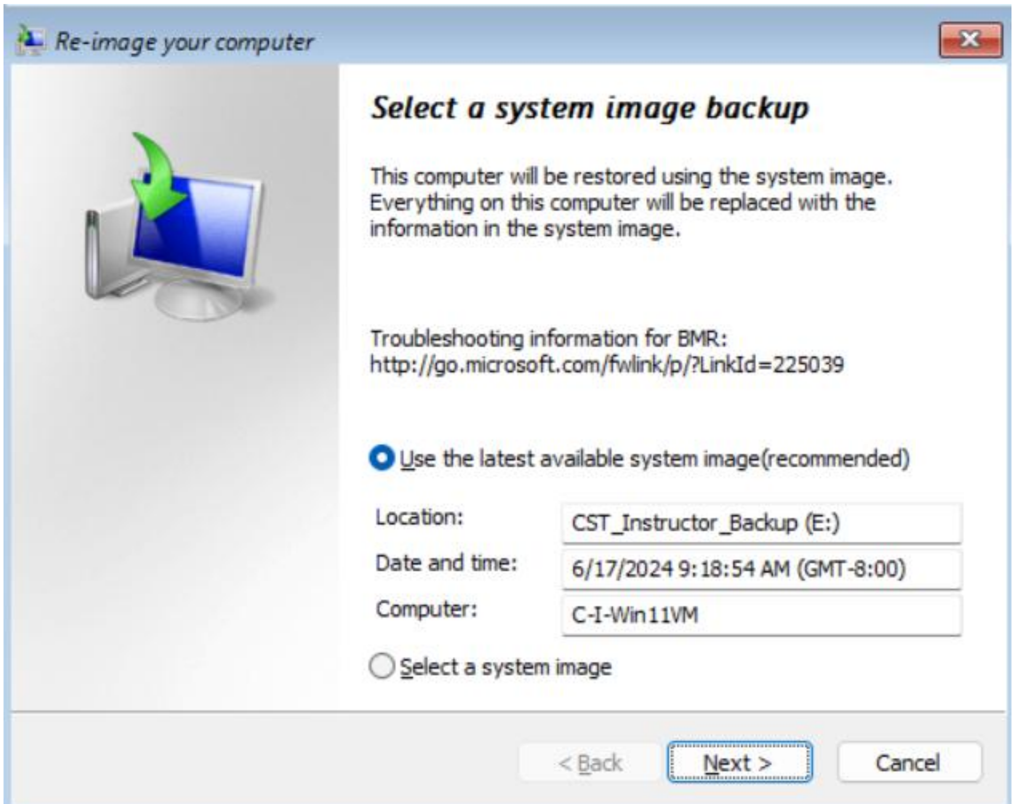
<<SS05>> Take a screenshot of the successful restore.



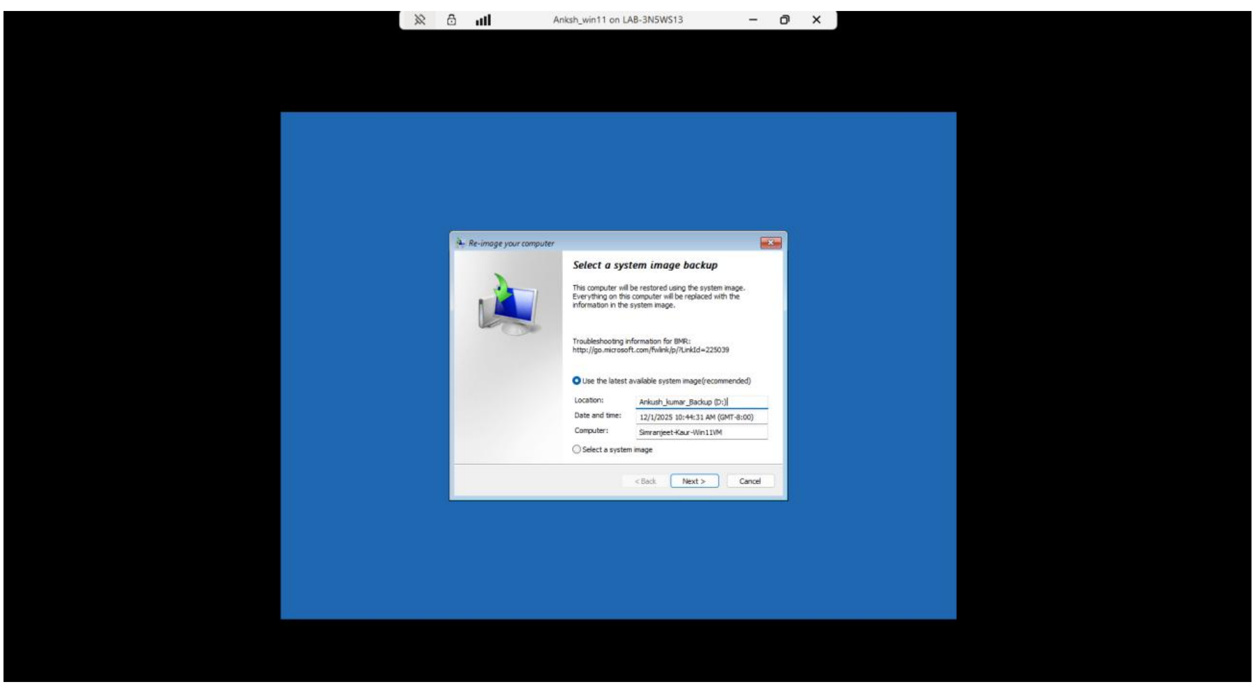
The restoration process can be used to restore files in case of loss of damage.

Restore from a system image

1. Reboot the PC into the recovery mode, (holding the Shift key) at the log on screen and restart.
2. Navigate to *Troubleshoot->Advanced options->See more recovery options->System Image Recovery.l*



<<SS06>> Take a screenshot of the reimage screen showing your back drive name as shown above.



3. Use the backup we created and click *Next*.
4. Click *Next*.
5. Click *Finish*.
6. Click *Yes*.
7. Restart when finished.

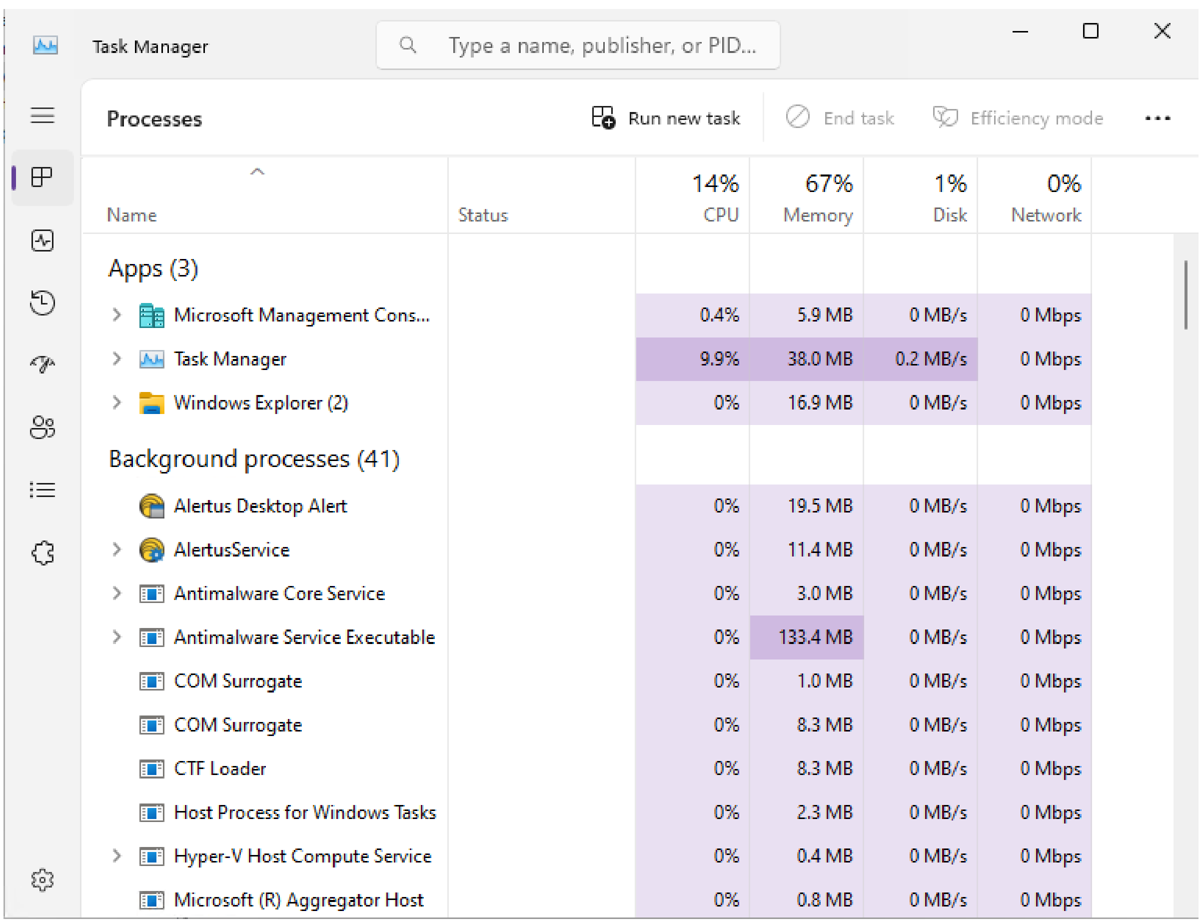
If this process failed, you will likely need to reinstall the PC from scratch.

B. Monitoring Windows

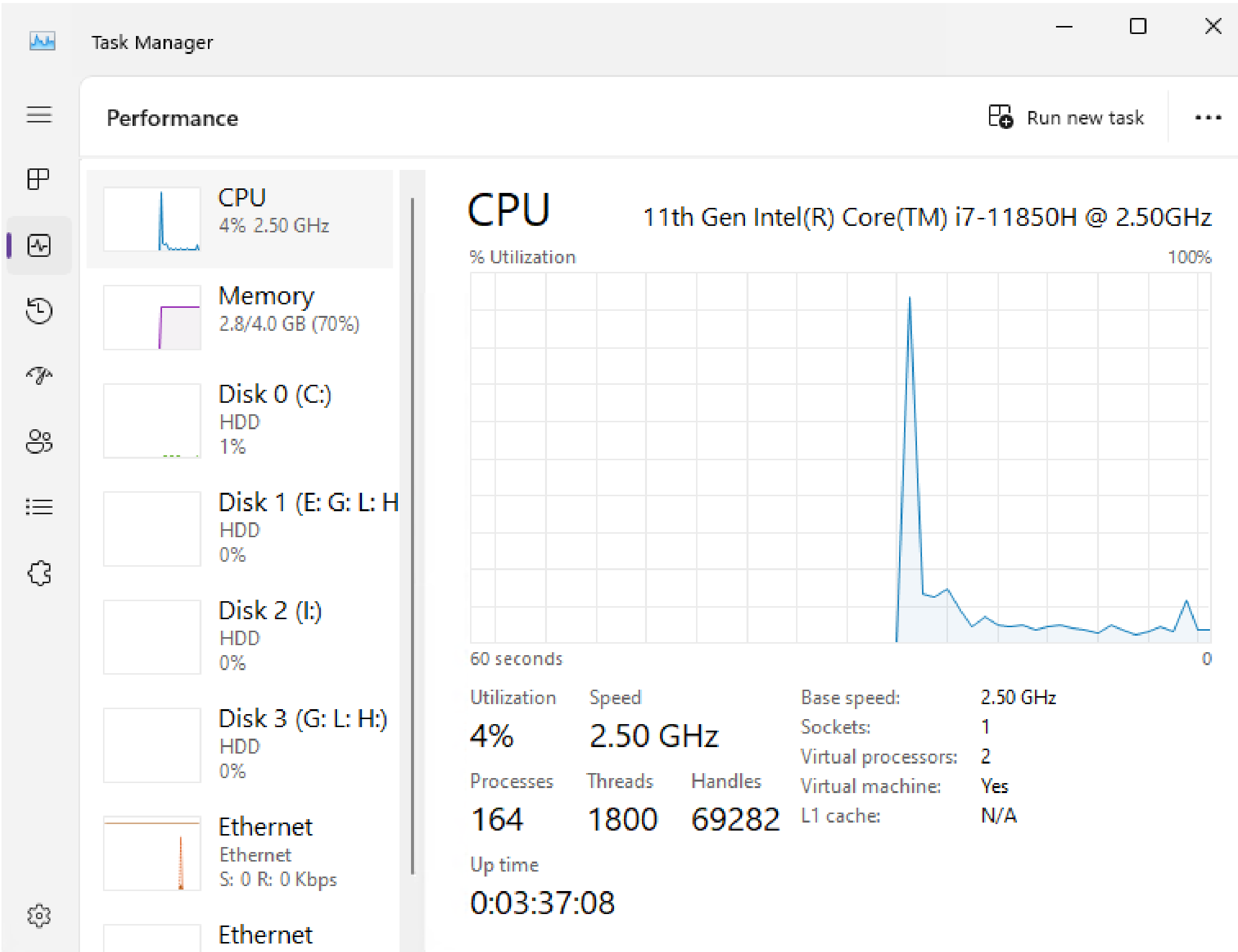
Task manager

A quick resource that allows you to view current process and a summary of resource usage

On most windows PC's ctr-alt-delete and select task manager. On newer Windows versions, right click the windows button and select Task manager.



Active performance can show if any resources are overallocated and is useful in initial debugging.



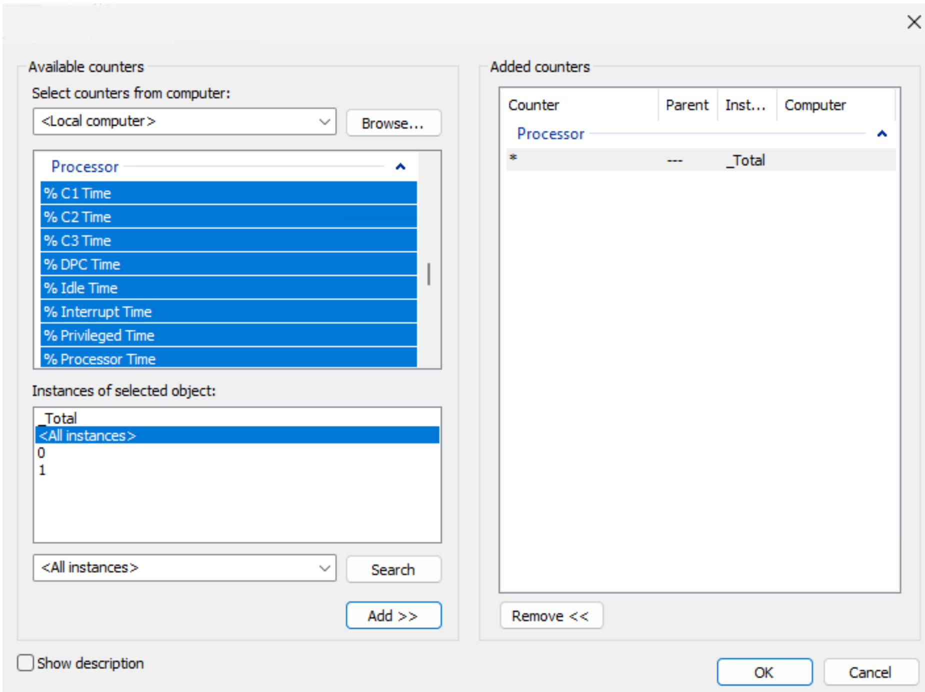
Performance monitor

1. Boot and log into Windows 10.
2. Search performance or navigate to *Control Panel->Adninistrative Tools->Performance Monitor.*
3. Open the Performance Monitor.

This tool allows you to monitor specific resources and create data collector sets of predefine resource groups.

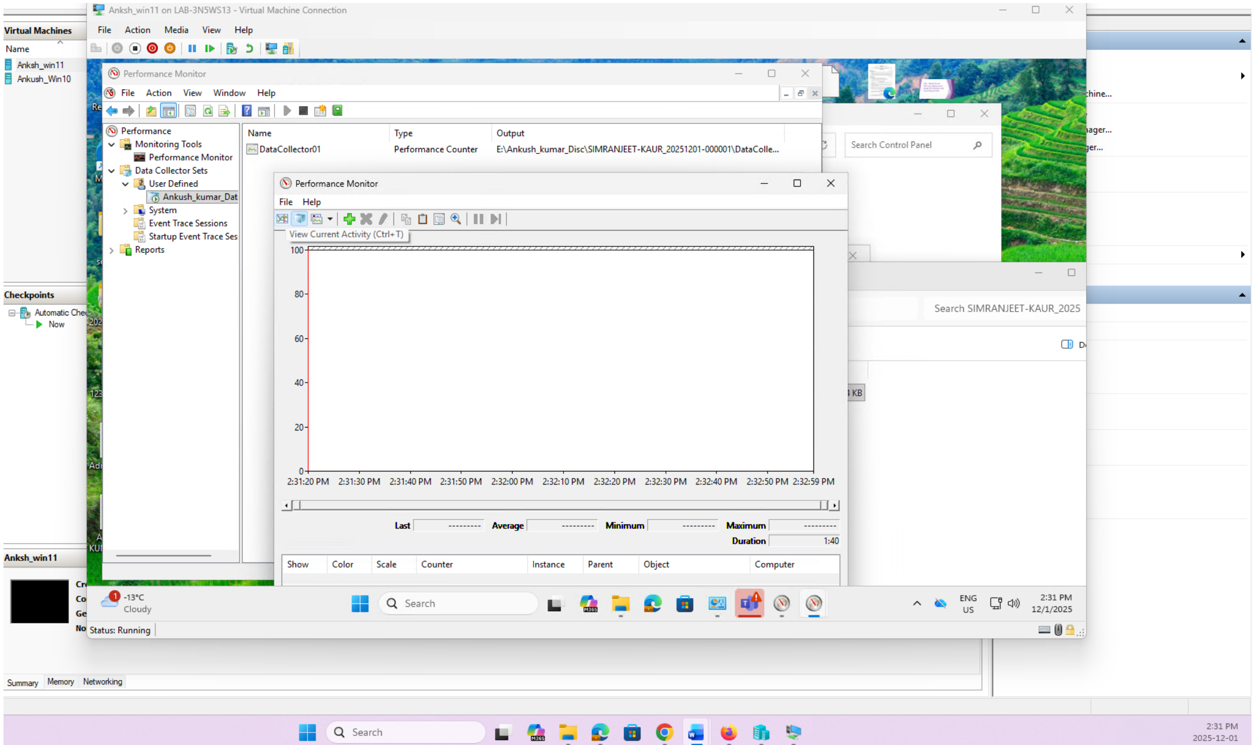
1. Open *Data Collector Sets->User Defined*.
2. Select *Action->New->Data Collector set*.
3. Name the set ***FirstName_LastName_DataSet***
4. Select Create Manually (Advanced) and click *Next*.
5. Select Create Data Logs, check Performance Counter then select *Next*.

6. Click *Add…*
7. Find Processor and select all counters.
8. Add to the counters.



9. Click *OK*.
10. Click *Next*.
11. Change the path to save it to your ***FirstName_LastName_Disc***
12. Run as default and select *Start this data collector set now*.
13. Click *Finish*.
14. Let it run for at least 15 seconds.
15. Navigate to your FirstName_LastName_Disc and open the folder that was created.
16. Open the performance log file.

<<SS07>> Take a screenshot of the performance log.

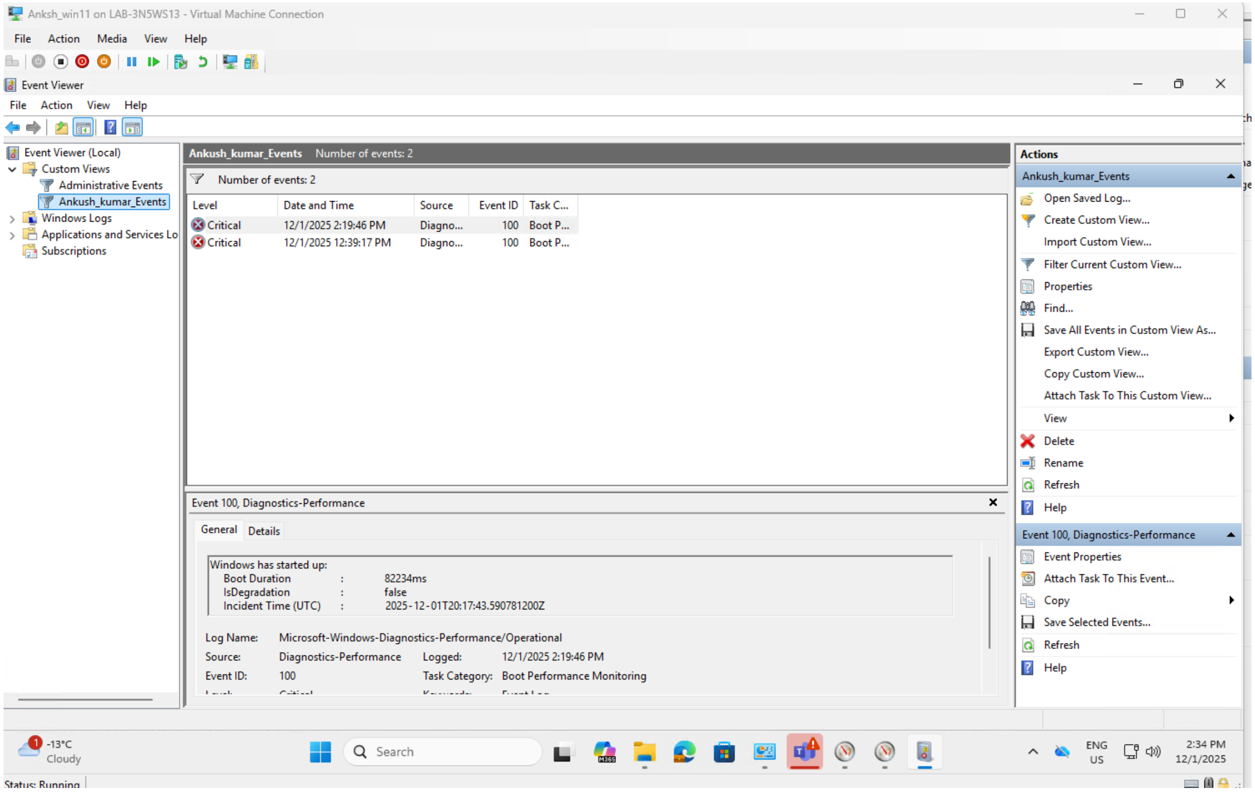


Event Viewer

Event Viewer allows you to see logs that Windows generates under certain conditions. We saw some in lab 2B when we viewed log on events.

1. Open event view by either searching Event Viewer or *Control Panel->Administrative Tools ->Event Viewer*. Event viewer can also be added to a view in the MMC.
2. Create a custom view for the event viewer by right clicking Custom views and selecting Create Custom View.
3. Create a view with the following parameters:
    a. Critical events only
    b. Logged in the last 24 hours
    c. By Log for Application and services logs
    d. By all Keywords
    e. For all users
4. Click *OK* when done.
5. Name the view ***FirstName_LastName_Events***
6. Save it in the custom views.

<<SS08>> Take a screenshot of the event viewer.



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